

BATCH 2 — EXPERIMENTS 4, 5, 6

These are taken from the uploaded experiment sheets, including the code visible in the screenshots. MATLAB-exp5 MATLAB-exp6



EXPERIMENT 4 — MATLAB

Exp. 4 has **Ex. 1 + Ex. 2 Part 1 + Ex. 2 Part 2**. MATLAB-exp4

Ex. 1 — Scalar, Vector & Matrix Functions

Copy into **exp4ann.m**

```
clc;  
clear;
```

```
A = [1 2 3;  
     4 5 6;  
     7 8 9];
```

```
v = [2 4 6 8];
```

```
disp('Scalar Functions');
```

```
disp(sin(pi/2));  
disp(cos(0));  
disp(tan(pi/4));  
disp(exp(1));  
disp(log(exp(1)));  
disp(abs(-10));  
disp(sqrt(25));  
disp(rem(17,5));  
disp(round(3.6));  
disp(floor(3.8));  
disp(ceil(3.2));
```

```
disp('Vector Functions');
```

```
disp(max(v));  
disp(min(v));  
disp(length(v));
```

```
disp(sort(v));
disp(sum(v));
disp(prod(v));
disp(median(v));
disp(mean(v));
disp(std(v));

disp('Matrix Functions');
```

```
disp(eye(3));
disp(zeros(3));
disp(ones(3));
disp(diag(A));
disp(tril(A));
disp(triu(A));
disp(rand(3));
```

```
disp(size(A));
disp(det(A));
disp(rank(A));
disp(rref(A));
disp(eig(A));
disp(poly(A));
```

Ex. 2 — Part 1

This solves the simultaneous equations using matrix division.

Copy into `exp4ann.m`

```
clc;
clear;
```

```
A = [4 -2 6;
     2 8 2;
     6 10 3];
```

```
B = [8;
     4;
     0];
```

```
X = A \ B;
```

```
disp('Solution of the equations');
```

```
disp('x =');  
disp(X(1));
```

```
disp('y =');  
disp(X(2));
```

```
disp('z =');  
disp(X(3));
```

Expected approximate result:

```
x =  
-1.80
```

```
y =  
0.29
```

```
z =  
2.63
```

Ex. 2 — Part 2

Dot product using 3 methods

```
clc;  
clear;
```

```
u = [-3 8 -2];  
v = [6.5 -5 -4];
```

```
% (a) Element-by-element using sum  
dot1 = sum(u .* v);
```

```
% (b) Matrix multiplication  
dot2 = u * v';
```

```
% (c) Built-in function  
dot3 = dot(u,v);
```

```
disp('(a) Dot Product using sum');  
disp(dot1);
```

```
disp('(b) Dot Product using matrix multiplication');  
disp(dot2);
```

```
disp('(c) Dot Product using dot function');
```

```
disp(dot3);
```

All three give:

```
-51.50
```



EXPERIMENT 5 — MATLAB

Your Exp. 5 has **5 exercises**. MATLAB-exp5 MATLAB-exp5

Ex. 1 — Function and Its Derivatives

```
clc;
```

```
clear;
```

```
close all;
```

```
x = -2:0.1:4;
```

```
y = 3*x.^3 - 26*x + 10;
```

```
dy = 9*x.^2 - 26;
```

```
d2y = 18*x;
```

```
plot(x,y,'b','LineWidth',2);
```

```
hold on;
```

```
plot(x,dy,'r','LineWidth',2);
```

```
plot(x,d2y,'g','LineWidth',2);
```

```
title('Function and its Derivatives');
```

```
xlabel('x');
```

```
ylabel('y');
```

```
legend('y = 3x^3 - 26x + 10', ...
```

```
    'First Derivative', ...
```

```
    'Second Derivative');
```

```
grid on;
```

Ex. 2 — Plot of Function

```
clc;
```

```

clear;
close all;

x = -3:0.1:5;

y = ((x + 5).^2) ./ (4 + 3*x.^2);

plot(x,y,'b','LineWidth',2);

title('Plot of f(x)');

xlabel('x');
ylabel('f(x)');

grid on;

```

Ex. 3 — VAB versus R1 and R2

```

clc;
clear;
close all;

V = 12;

R3 = 250;
R4 = 250;

% Plot (a)

R2 = 120;
R1 = 0:1:500;

vAB1 = V*((R2./(R1+R2)) - (R4/(R3+R4)));

subplot(2,1,1);

plot(R1,vAB1,'b','LineWidth',2);

title('V_{AB} versus R_1');

xlabel('R_1 (Omega)');
ylabel('V_{AB} (V)');

grid on;

% Plot (b)

```

```

R1 = 120;
R2 = 0:1:500;

vAB2 = V*((R2./(R1+R2)) - (R4/(R3+R4)));

subplot(2,1,2);

plot(R2,vAB2,'r','LineWidth',2);

title('V_{AB} versus R_2');

xlabel('R_2 (Omega)');
ylabel('V_{AB} (V)');

grid on;

```

Ex. 4 — Cosine Signal

```

clc;
clear;
close all;

x = 0:1:100;

y = cos(x);

plot(x,y,'--rs', ...
     'LineWidth',1.5, ...
     'MarkerSize',5, ...
     'MarkerEdgeColor','b', ...
     'MarkerFaceColor','c');

title('Cosine Signal');

xlabel('Time (s)');
ylabel('Amplitude (m)');

set(gca,'Color','y');

grid on;

```

Ex. 5 — Nine Tone Signal

```

clc;

```

```

clear;
close all;

Fs = 8000;      % Sampling frequency
duration = 0.5; % Duration of each tone

freq = [659 622 659 622 659 523 440 523 440];

signal = [];

for i = 1:length(freq)

    t = 0:1/Fs:duration;

    tone = sin(2*pi*freq(i)*t);

    signal = [signal tone];

end

modifiedSignal = signal;

modifiedSignal(2:2:end) = modifiedSignal(2:2:end);

finalSignal = [signal modifiedSignal];

sound(finalSignal,Fs);

audiowrite('NineTone.wav',finalSignal,Fs);

figure;

spectrogram(finalSignal,256,200,256,Fs,'yaxis');

title('Spectrogram of Nine Tone Signal');

xlabel('Time (s)');
ylabel('Frequency (kHz)');

```

This one creates **NineTone.wav** in your current MATLAB folder.



EXPERIMENT 6 — MATLAB

Exp. 6 has **Ex. 1–5**. MATLAB-exp6 MATLAB-exp6

Ex. 1 — Signum Function

```
clc;
clear;
close all;

t = -10:0.1:10;

y = sign(t);

plot(t,y,'b','LineWidth',2);

title('Signum Function');

xlabel('Time');
ylabel('Amplitude');

grid on;
```

Ex. 2 — Exponential Growing Signal

```
clc;
clear;
close all;

t = 0:0.1:5;

y = exp(t);

plot(t,y,'r','LineWidth',2);

title('Exponential Growing Signal');

xlabel('Time');
ylabel('Amplitude');

grid on;
```

Ex. 3 — Exponential Decaying Signal

```
clc;
clear;
close all;
```

```
t = 0:0.1:5;

y = exp(-t);

plot(t,y,'b','LineWidth',2);

title('Exponential Decaying Signal');

xlabel('Time');
ylabel('Amplitude');

grid on;
```

Ex. 4 — Subtraction and Multiplication of Sinusoidal Signals

```
clc;
clear;
close all;

t = 0:0.01:2*pi;

x1 = sin(t);
x2 = sin(2*t);

subplot(2,1,1);

plot(t,x1-x2,'b','LineWidth',2);

title('Subtraction of Two Sinusoidal Signals');

xlabel('Time');
ylabel('Amplitude');

grid on;

subplot(2,1,2);

plot(t,x1.*x2,'r','LineWidth',2);

title('Multiplication of Two Sinusoidal Signals');

xlabel('Time');
ylabel('Amplitude');
```

grid on;

Ex. 5 — Fahrenheit to Celsius + Change in Area

This exercise uses a function file.

STEP 1 — Create **FtoC.m**

```
function C = FtoC(F)
```

```
C = (F - 32) * 5 / 9;
```

```
end
```

Important: save this file exactly as:

FtoC.m

STEP 2 — Create another script, e.g. **exp6_ex5.m**

```
clc;
```

```
clear;
```

```
alpha = 2e-6;
```

```
L = 4.5;
```

```
W = 2.5;
```

```
F1 = 40;
```

```
F2 = 92;
```

```
C1 = FtoC(F1);
```

```
C2 = FtoC(F2);
```

```
deltaT = C2 - C1;
```

```
deltaL = alpha * L * deltaT;
```

```
deltaW = alpha * W * deltaT;
```

```
deltaArea = (L * deltaW) + (W * deltaL);
```

```
disp('Temperature change (C)');
```

```
disp(deltaT);
```

```
disp('Change in Area (m^2)');
```

```
disp(deltaArea);
```

Run `exp6_ex5.m`

Make sure **both files are in the same MATLAB folder:**

FtoC.m

exp6_ex5.m

Then run:

```
exp6_ex5
```

Batch 2 done

Next is **BATCH 3** → **EXPERIMENT 7, 8, 9**. Those are much bigger, especially **Exp 7** and **Exp 8**, so I'll keep every exercise separate like this rather than dumping one giant script.